AES SHOW 2026 NASHVILLE · 161ST CONVENTION · EXPRESS PAPER

A Deterministic Multi-Agent Architecture for Voice-Controlled DAW Interaction

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Sunday 2026-11-01 · 10:30–11:00 · Volunteer Mezzanine 10 · Express Paper NPNRJW7MPR

The studio never got an interface

Three takes in. The vocal needs two more dB.
The options: stop playing — or say something.

THE INSTRUMENT

Both hands occupied.
Eyes on the performance.

Nothing in the signal path answers to a word.

THE DESK

Every control lives
one hand away.

Reaching for it costs the take.

Voice demos well. It has never been handed a session.

Why it failed: the stateless system

Every utterance started from zero.

01 No session awareness

02 No verification that anything happened

03 No memory of what it just did

A system you can’t trust with a session gets switched off.

The design constraints

01 / Headless.

02 / Non-generative — it assists; it writes nothing.

03 / Session-aware.

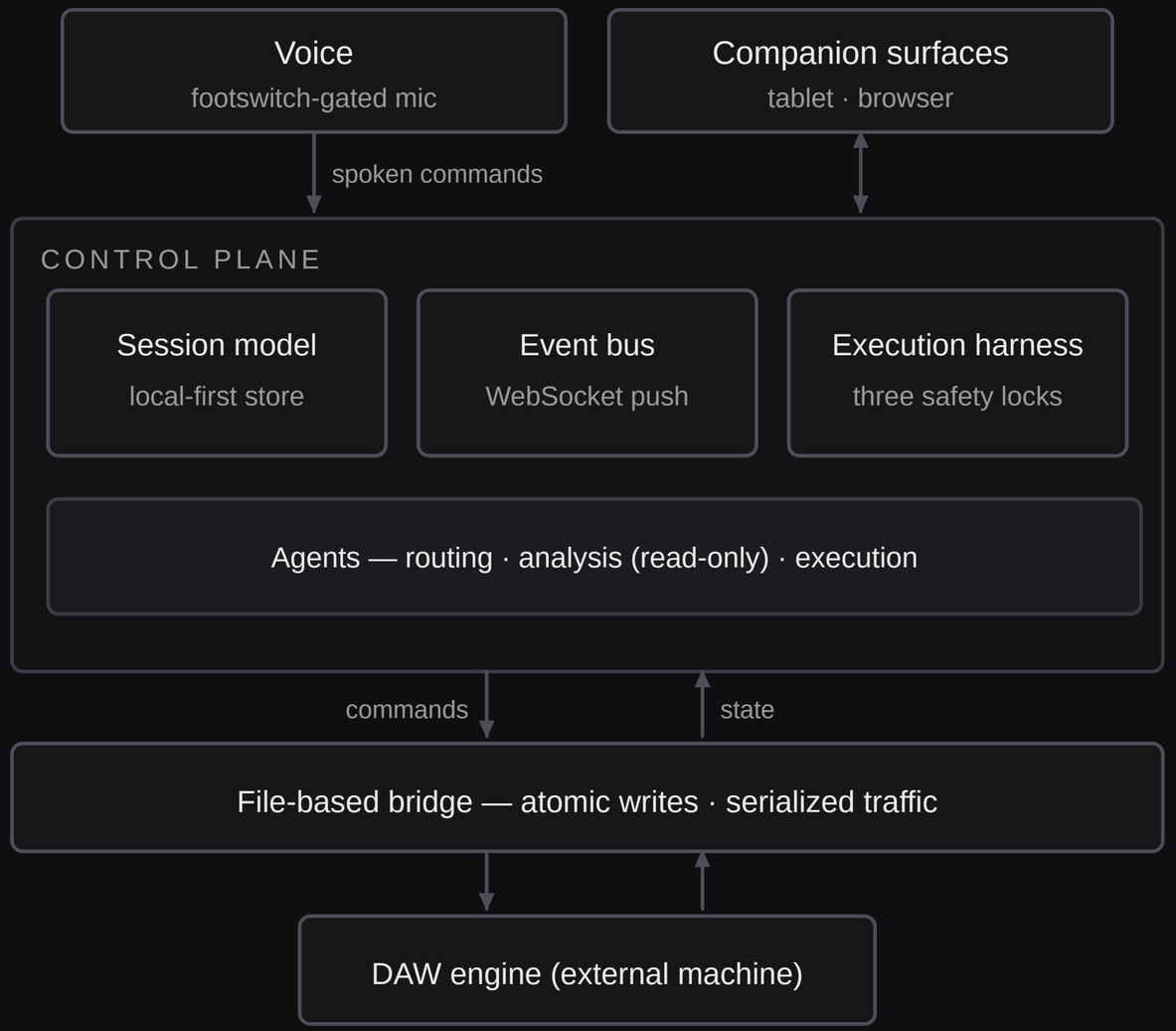
04 / Verifiable at every step.

Four constraints, set before any model was chosen — and never softened afterwards.

Architecture — three layers

Deterministic mechanical layer.
Constrained intelligence. Verification.

The model proposes.
The execution layer verifies.

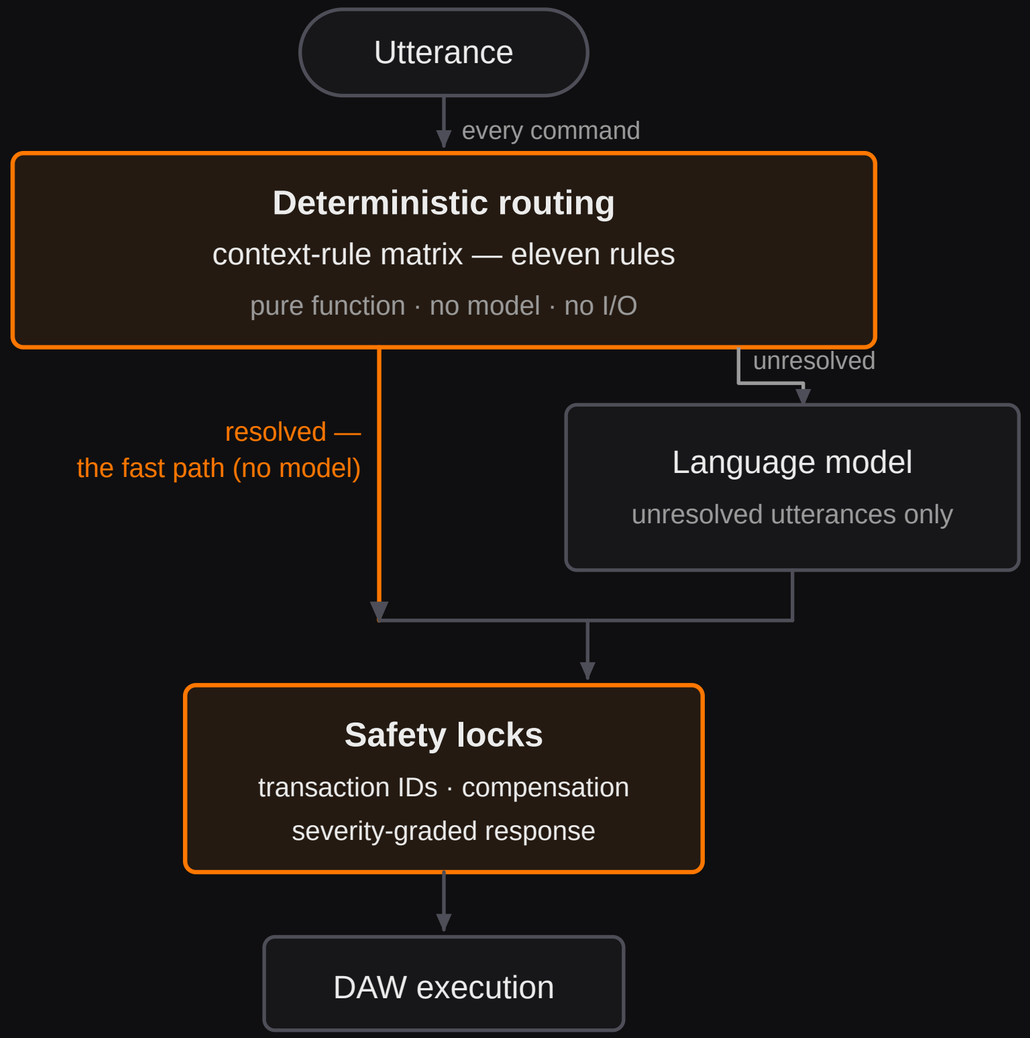


The routing layer is the spine

Eleven deterministic rules evaluate the session context — a pure function.

The model is the fallback, not the front door.

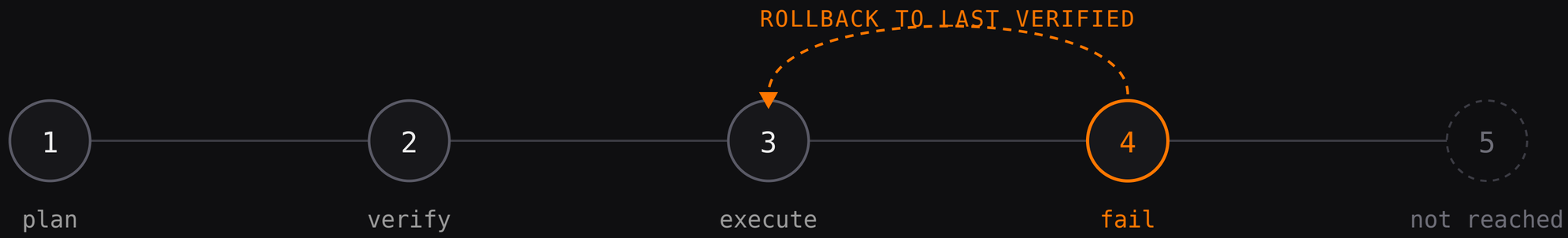
Meaning lives in context, not in the words.



The fast path never leaves the deterministic layer.

Every step is verified

Verify each step before the next.



On failure: roll back to the last verified step — not the beginning.

Most systems know whether the plan failed. This one knows where.

Failure, graded

TRANSIENT	Logged, silent, session continues.
PERMANENT	Rolled back, then named out loud: “I rolled that back.”
CATASTROPHIC	The console locks out until the user takes over.

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CONSOLE LOCKOUT  commands refused → user takes control
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A failure the session survives — by design.

LIVE

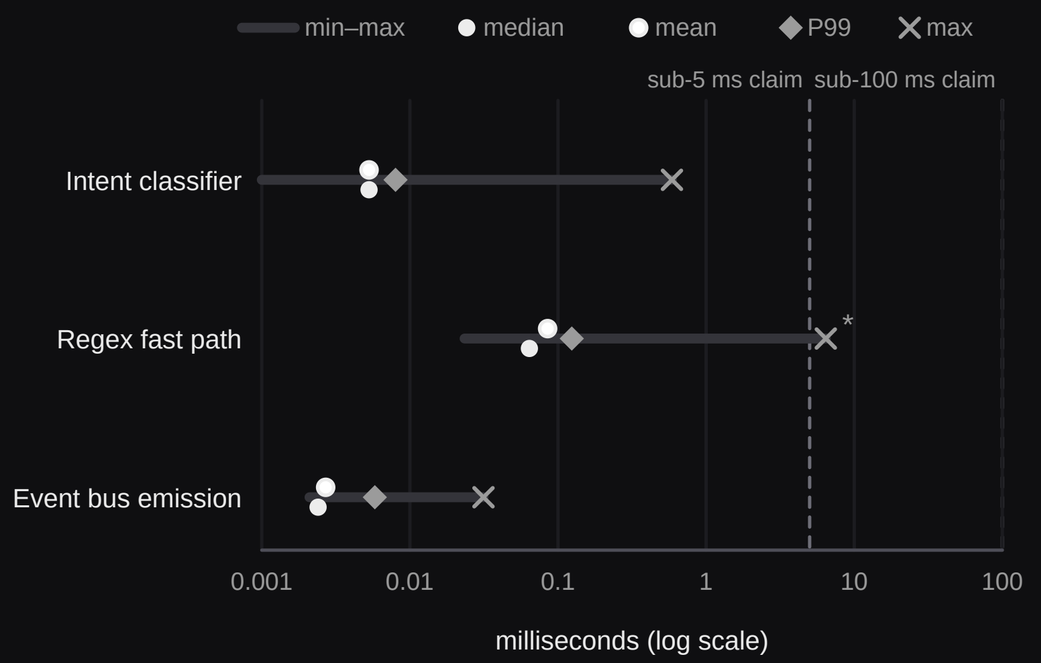
The rollback beat

OPEN COMMAND REFUSE

Measured

INTENT CLASSIFIER	P99 0.008 ms
REGEX FAST PATH	P99 0.12 ms
EVENT BUS EMISSION	P99 0.006 ms
LOCK TESTS	12

1000-command run, per subsystem — P99 shown; min/max in Figure 3.



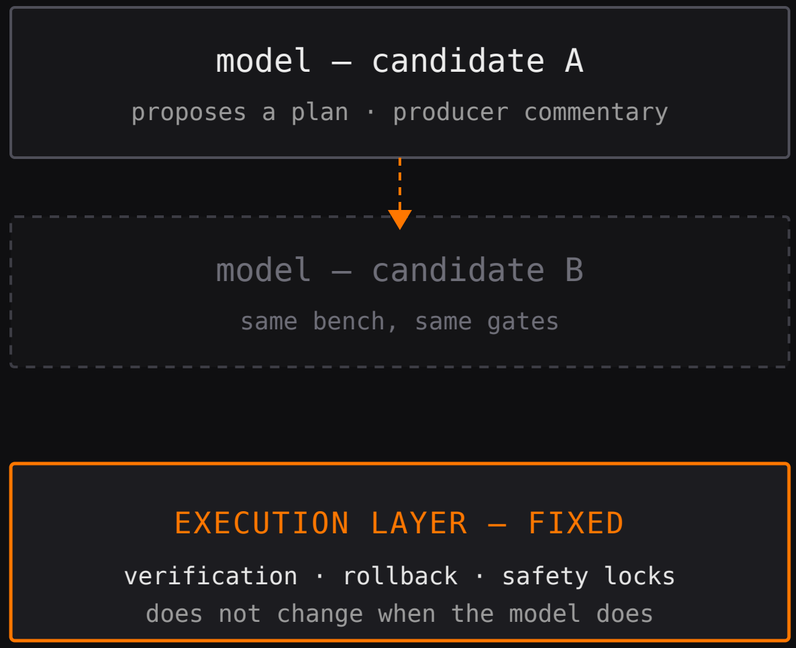
* first-invocation outlier (JIT warmup); aggregates from the 1000-command run.

The model is a part. Not the instrument.

Four gates, declared before measurement. Two jobs for models — everything else runs without them.

A slow response is a delay.
No response is a lost take.

no cloud dependency in the session path



The missing layer was trust.

Determinism where the session depends on it.
Models where judgment is needed.

Assist — never generate.

What would you build differently if a failure never cost the take?

Thank you.

Backups — appendix, not shown unless needed

A1	The eleven routing rules, listed
A2	Step-verification mechanics — ledger schema, timing
A3	Routing fallback decision table
A4	Eval methodology and pre-declared gates
A5	References — the paper
–	Available on request